

Impact Statement — AgriAdviser-NG

AgriAdviser-NG aims to give smallholder rice and maize farmers in Nigeria fast, localized agronomic guidance over USSD, SMS, and WhatsApp, using a small language model fine-tuned on extension bulletins from IITA, AfricaRice, NAERLS, and state Agricultural Development Programmes. Its most direct benefit is timeliness: a farmer noticing early pest damage or unusual leaf discoloration today typically waits days for an extension officer visit, by which point a treatable problem can become a lost season. Reducing that gap has concrete economic value, a correctly identified fall armyworm infestation caught within a week of onset is far cheaper to manage than one caught a month later, and for a smallholder household, that difference can represent a meaningful share of annual income. Beyond individual farms, faster access to accurate guidance could modestly ease pressure on Nigeria's limited pool of human extension officers, who are typically responsible for far more farmers than they can visit regularly, allowing their time to be redirected toward cases the model cannot resolve.

The project also carries real risks that scale differently than a technical demo suggests. A model that is wrong with confidence is more dangerous than one that admits uncertainty, because farmers may act on its advice with limited ability to independently verify it, which is why abstention, not answer coverage, is the design's central safeguard, and why every reference answer in the dataset must be traceable to a real source before it is trusted. There is also a fairness risk in who the system currently serves well: it is built for literate, phone-owning farmers who type in English or Pidgin, which by construction underserves non-literate farmers and, per well-documented patterns in agricultural extension data, likely underrepresents women farmers' questions and priorities unless deliberately corrected. Left unaddressed, a tool meant to reduce inequality in access to expertise could instead concentrate its benefit among farmers already best positioned to seek help elsewhere, widening rather than narrowing an existing gap.

There are autonomy and dignity considerations too. The system is explicitly scoped to augment, not replace, human extension officers, partly because agricultural decisions carry livelihood stakes that warrant a human accountable party, and partly because an over-trusted automated advisor risks eroding farmers' own accumulated local knowledge rather than complementing it. Environmentally, the offline-first, CPU-only deployment target is a deliberate choice to avoid assuming energy or connectivity infrastructure this population cannot reliably access, though at scale, hardware distribution and device lifecycle impacts would need separate assessment not yet undertaken here.

Significant uncertainty remains, and should be acknowledged rather than smoothed over. The dataset's current reliance on synthetic Pidgin phrasing rather than authentic farmer speech means the model's real-world comprehension of how farmers actually ask questions is untested; NAERLS and IITA source coverage across Nigeria's diverse agro-ecological zones has not yet been audited for gaps; and the abstention mechanism's real-world reliability under ambiguous, multi-symptom questions has not been evaluated outside controlled testing. All three should be resolved with real pilot data before any claim of broad reliability is warranted.